Do not write outside the box

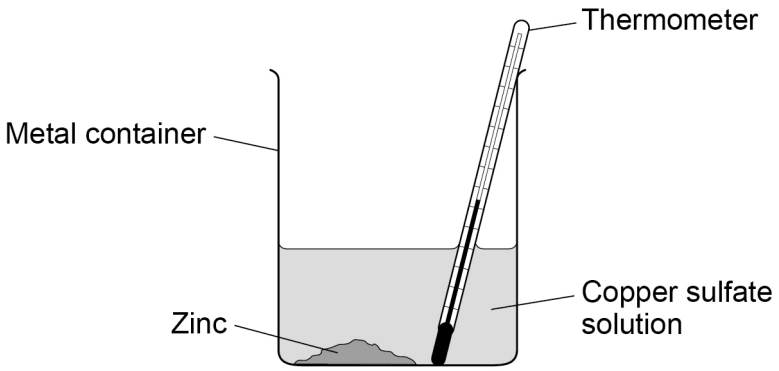
0 5 A student investigated the change in temperature when different masses of zinc were added to copper sulfate solution.

This is the method used.

1. Measure the volume of copper sulfate solution using a measuring cylinder.
2. Pour the copper sulfate solution into a metal container.
3. Add 2 g of zinc.
4. Measure the temperature of the solution.
5. Repeat steps 1 to 4 with different masses of zinc.

Figure 7 shows the apparatus.

Figure 7



0 5 . 1 Give **three** improvements to the investigation to make the results more accurate.

[3 marks]

1 _____

_____

2 _____

_____

3 _____

_____

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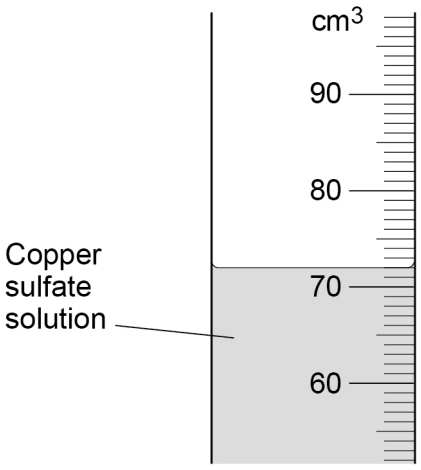
0 5 . 2 **Figure 8** shows part of the measuring cylinder.

Figure 8



What is the volume of copper sulfate solution in **Figure 8**?

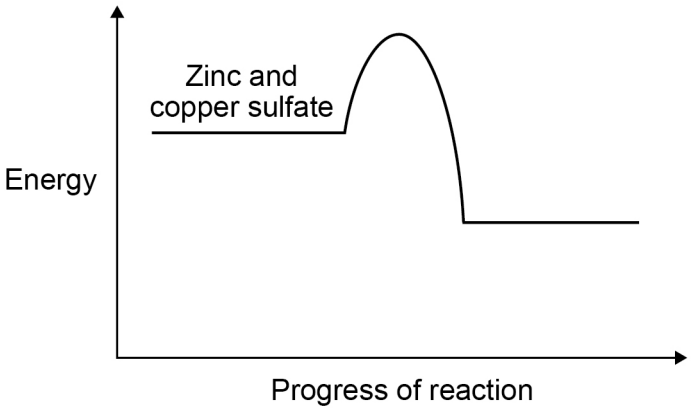
[1 mark]

Volume = _____ cm^3

0 5 . 3 When zinc was added to copper sulfate solution the temperature increased.

Figure 9 shows the reaction profile.

Figure 9



What type of reaction is shown in **Figure 9**?

[1 mark]

Tick (✓) **one** box.

Endothermic ☐ Exothermic ☐ Neutralisation ☐

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Figure 10 shows the results.
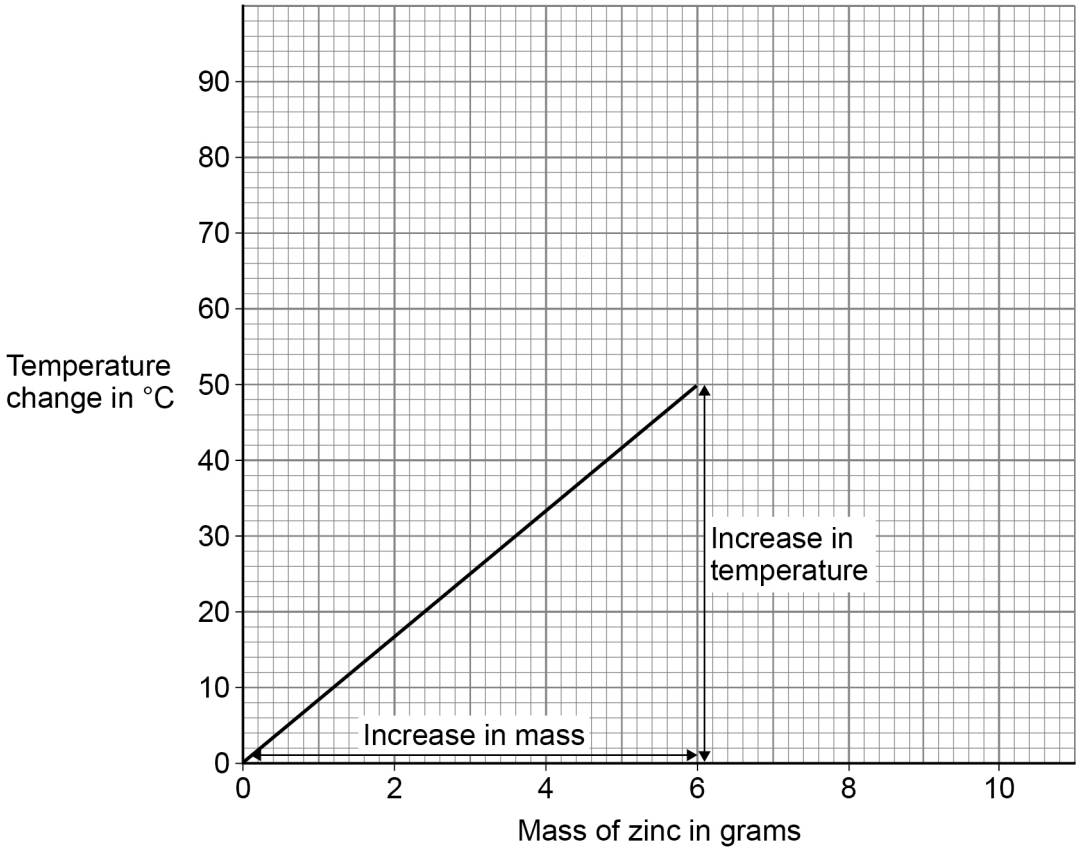
Figure 10